

FULL STACK COMMUNITY

BIZ HACK'26

ONE-DAY FULL STACK & SOFTWARE DEVELOPMENT HACKATHON

SOFTWARE REQUIREMENTS SPECIFICATION

Team Name	CODE BLENDERS
Team ID	FSSD-071
Problem Statement No.	PS78
Problem Statement Title	Scholarship Monitoring System
Team Leader	Jay Dinakar.R
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Team Members

#	Name	Role
1	Jay Dinakar.R	Frontend and Backend Development
2	Selvasanjiv.G	Database Configuration
3	Sanjay.P	Testing and Deployment
4		

01 PROBLEM UNDERSTANDING

In 3–5 lines, explain the problem in your own words. Who is affected, what breaks down today, and why does it matter?

Traditional scholarship administration across higher education institutions suffers from manual bureaucracy, fabricated documentation (photoshopped income certificates, counterfeit caste attestation, and doctored grade sheets), lack of quota transparency, and opaque disbursement delays averaging 6 to 14 months.

Students lack real-time visibility into their application stages, while committee administrators struggle to verify statutory credentials across heterogeneous portals. This systemic vulnerability leads to intermediary leakage, quota exhaustion by ineligible candidates, and denial of opportunities to genuinely deserving high-merit scholars.

02 PROPOSED SOLUTION

Describe what your team will build and how it solves the problem above. One short paragraph is enough — save the detail for Section 03.

Our team built an enterprise-grade, tamper-proof Scholarship Monitoring & Direct Benefit Transfer (DBT) System for Bannari Amman Institute of Technology scholars. The platform unifies automated Indian statutory document forensics (UIDAI Verhoeff check-digit validation, Tahsildar seal heuristics, and a 6-document depository via DigiLocker NAD) with a public immutable SHA-256 cryptographic audit ledger.

Key capabilities include Explainable AI Eligibility Matching with strict CGPA enforcement (strictly blocking ineligible applicants before entry), atomic quota/slot tracking, official vector PDF award letters signed by Dean Jay Dinakar R, a 1-click multi-file statutory audit ZIP pack exporter, and an interactive onboarding guide tour (?).

03 CORE FUNCTIONAL REQUIREMENTS

List only the features you plan to actually implement during the hackathon. Priority: High / Medium / Low.

Requirement / Feature	Priority
Role-Based Authentication & Session Guard (Student vs Admin)	High
6-Document Statutory Depository & DigiLocker Sync (Aadhaar, Marksheet, Bonafide, Income, Caste, Passbook)	High
Instant High-Resolution Specimen Proof Previews (Zero-Delay SVG/PNG Modals)	High
Dedicated Replacement Pickers with Slot Locking (uploadLockedDocType)	High

Explainable AI Match & Strict CGPA Enforcement Rule (< minGpa blocks submission)	High
Institutional Bonafide Verification & 100% Profile Completion Transition (All Clear)	High
Admin Scheme Management & Live Quota Slot Tracking (Create, Edit, Archive Schemes)	High
Direct Benefit Transfer (DBT) Simulation via NPCI PFMS Bridge (State Bank of India IFSC)	High
Bannari Amman Institute of Technology Official Award Letter Generation (PDF & PNG)	High
Public SHA-256 Cryptographic Audit Ledger with Block Hash Search	High
Statutory Audit Pack ZIP Export (CSV Summary, JSON Ledger, TXT Report)	High
Interactive Platform Walkthrough Guide Modal (?) with Step-by-Step Tour	Medium

04 USER ROLES

List each actor/role in the system and what they can do.

User / Role	Main Responsibilities
Admin	Manage the complete scholarship process: create scholarships, define eligibility criteria, set award limits, review student applications, approve or reject applications, and monitor available slots.
Student	Create an account/login, view available scholarships, check eligibility, submit applications, and track application status from submission to final decision.
System	Validate application actions, maintain application status, automatically update remaining award slots after approval, and close scholarships when all awards are allocated.

05 SYSTEM WORKFLOW

Show the main flow of the application — a simple flowchart or numbered sequence of screens/actions is recommended over prose.

1. Student Authentication & DigiLocker Sync:

Student logs in using Roll No or Email; system auto-links profile and fetches 6 statutory credentials from DigiLocker NAD.

2. Automated Document Forensics & Verification:

Certificates undergo client-side SHA-256 hashing, UIDAI Verhoeff check-digit scans, and institutional seal validation. Authentic specimen proofs are immediately inspectable.

3. Scheme Discovery & Explainable Eligibility Matching:
System evaluates CGPA, family income, and engineering branch, computing match scores (up to 98%). If a student's CGPA is below the minimum threshold, application buttons are disabled with 'You Are Not Eligible'.
4. Smart Multi-Step Application Flow:
Pre-populates 85% of application data across 6 structured steps, concluding with statutory applicant declaration and Merkle ledger anchoring.
5. Committee Review & DBT Bank Disbursal:
Admin evaluates dossiers, verifies statutory attachments, approves quota deductions, and simulates Direct Benefit Transfer (DBT) grants via the NPCI PFMS bank gateway.
6. Official Award Letters & Compliance Export:
Recipients download high-resolution vector PDF & PNG award letters signed by Dean Jay Dinakar R. Auditors export complete compliance packs (ZIP) containing CSV data, JSON ledgers, and audit logs.

06 TECHNOLOGY STACK

Default MERN layout shown below — edit any row to match your actual stack.

Layer / Component	Technology / Tool
Frontend Framework	Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS 4
UI & Interactivity	Motion (Framer Motion), Lucide React, Canvas Confetti, Generative UI
Backend & Serverless	Next.js Server Actions, API Route Handlers, Edge Runtime
Database & Persistence	Supabase PostgreSQL (Cloud) + Encrypted Local Storage Engine
File & Specimen CDN	Cloudinary CDN + Local Vector SVG & PNG Specimen Depository
Forensics & Security	UIDAI Verhoeff Dihedral D5 Checksum Engine, Client SHA-256 Hashing
Document Generation	jsPDF 4.2 (Vector PDF Award Letters & SRS), JSZip 3.10 (Audit Packs)
Hosting & Infrastructure	Vercel Edge Cloud Deployment + GitHub Version Control
Statutory Integrations	DigiLocker NAD API Standard, NPCI DBT APBS Bridge, PFMS Guidelines

07 DATABASE / DATA REQUIREMENTS

List the main entities/collections and the key fields each one stores.

Entity / Table	Key Data / Fields
Users / StudentProfiles	id, fullName, email, rollNo, department, gpa, annualIncome, quotaCategory, dbtBankName, dbtAccountNo, dbtIfscCode, avatarUrl

Scholarships	id, title, description, category, awardAmount, totalSlots, remainingSlots, deadline, minGpa, maxIncome, eligibleDepartments, status, createdBy
Applications	id, scholarshipId, scholarshipTitle, studentId, studentName, studentEmail, studentDepartment, studentGpa, annualIncome, essay, status, reviewerName, reviewNotes, sha256Hash, createdAt, updatedAt
VaultDocuments	id, studentId, title, issuer, fileName, fileSize, validity, isVerified, sha256Hash, fileUrl, fraudRiskScore, verdict, extractedIdentifier
LedgerBlocks	blockNumber, timestamp, eventType, recordId, previousHash, blockHash, merkleRoot, actor, status, metadata

08 NON-FUNCTIONAL REQUIREMENTS

Brief expectations around quality attributes — keep each to one line.

Category	Expectation
Performance	Core pages and actions should respond quickly during normal hackathon demonstration and usage.
Security	Passwords must be securely hashed; authentication and role-based protected routes must be used.
Usability	Clean, responsive interface with simple navigation and clear scholarship/application status.
Availability	The deployed application should remain accessible throughout the judging and demonstration period.

09 MVP SCOPE

Separate what's essential to demo from what's a stretch goal if time allows.

Must-Have Features	Optional / If Time Permits
• Role-Based Dual Authentication (Student Scholar & Committee Chair)	• Interactive Platform Tour & Navigation Walkthrough (?) with Previous/Next controls

• DigiLocker Statutory Document Depository for all 6 required certificates • Real-time Indian anti-fraud engine (UIDAI Verhoeff Dihedral D5 check-digits) • Instant specimen proof previews for Aadhaar, Marksheet, Bonafide, Income, Caste, Passbook • Strict CGPA enforcement blocking application when student CGPA < minGpa • Dynamic transition of profile completion from 90% to 100% upon bonafide attestation • Admin scholarship creation, live editing, and quota slot atomic updates • Bannari Amman Institute of Technology Official Award Letter generation (Vector PDF & PNG) • Direct Benefit Transfer (DBT) bank payment simulation via NPCI IFSC bridge • Public SHA-256 Cryptographic Audit Ledger with hash verification • 1-Click Audit Pack ZIP bundle export (CSV summary, JSON ledger, TXT report)	• WhatsApp status alert notification simulation • Multi-criteria smart filtering and search across 150+ national programs • Dynamic persona switcher between high-merit and need-based applicants
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10 EXPECTED OUTPUT

In 2–3 lines, describe what the final working demo should show.

A fully functional, zero-leakage Scholarship Monitoring & Direct Benefit Transfer (DBT) web application running live on Vercel and localhost:3000.

The demonstration showcases automated DigiLocker ingestion, instant high-res specimen proof previews, strict CGPA enforcement blocking ineligible candidates, real-time quota slot management, live committee approval with DBT bank disbursement, downloadable Bannari Amman Institute of Technology award letters (PDF/PNG), interactive platform walkthrough tour (?), and a 1-click downloadable cryptographic audit ZIP pack.

TEAM CONFIRMATION

We confirm that this document represents our proposed implementation for the selected BIZ HACK'26 problem statement.

JAY DINAKAR R

Team Leader Signature: _____

23.09.2026

Date: _____